

# Nils Lukas

Assistant Professor • MBZUAI • Abu Dhabi, UAE

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## Research Interests

Design secure and private Machine Learning systems in the presence of untrustworthy

1. **Providers:** Confidential computing via Homomorphic Encryption & Secret Sharing.
2. **Data:** Mitigate data poisoning during training & prompt injection during inference.
3. **Models:** Protect training data privacy through PII scrubbing & differential privacy.
4. **Users:** Control misuse by detecting generated (mis)information with watermarking.

## Education

**University of Waterloo, Canada** 2019 – 02/2024  
Ph.D. in Computer Science  
• Advisor: [Florian Kerschbaum](#)  
• Thesis: Analyzing Threats of Large-Scale Machine Learning Systems  
• Thesis Awards: [Top Mathematics Doctoral Prize](#) & [Alumni Gold Medal](#)

**RWTH-Aachen, Germany**  
M.Sc. in Computer Science (*w/Distinction*) 2016 – 2018  
B.Sc. in Computer Science 10/2012 – 2016

## Honors & Awards

[Amazon Research Award](#) [100,000 USD] 2025  
First Place at the NeurIPS'24 Watermarking Competition [4,400 USD] 2024  
First Place at DGE Elite Hackathon, GITEX'24 [10,900 USD] 2024  
Top Mathematics Doctoral Thesis, University of Waterloo [1,080 USD] 2024  
Alumni Gold Medal, One PhD Award Yearly, University of Waterloo 2024  
Best Poster Award, Sponsored by David R. Cheriton [220 USD] 2023  
Distinguished Contribution Award, Microsoft MLADS conference 2023  
David R. Cheriton Scholarship, University of Waterloo [14,400 USD] 2022, 2023  
Outstanding Reviewer, ICML'22 2022  
Best Poster Award, Sponsored by Rogers [720 USD] 2019

## Research Grants

**Awarded**

**[Etihad Airways] Conversational Booking Agents** **\$450,000** · 2025  
PI: Nils Lukas · Co-PIs: Salem Lahlou, Alham Fikri, Martin Takáč, Mingming Gong

**[United Al-Saqer Group] Privacy-preserving Brain–Computer Interfaces** **\$136,000** · 2025  
PI: [Abdulrahman Mahmoud](#) · Co-PIs: Nils Lukas, Elizabeth Churchill

**[Amazon Research Awards – Special Call] AdvSim2Real: Robustifying Agents in Simulated Adversarial Environments** **\$100,000** · 2025  
PI: Nils Lukas · Co-PI: [Praneeth Vepakomma](#)

**Selected**

**[TII Funding] GFlowNets for Fuzzing of Agentic Applications** **\$136,000** · 2025  
PIs: Salem Lahlou & Nils Lukas

## Conference Publications

<sup>22</sup>  
**[CVPR'26]** [RAVEN: Erasing Invisible Watermarks via Novel View Synthesis](#)  
Fahad Shamshad, Nils Lukas, Karthik Nandakumar. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.  
 **Oral (top 0.8%)**

<sup>21</sup>  
**[ICML'26]** [Collaborative Threshold Watermarking](#)  
Tameem Bakr, Anish Ambreth, Nils Lukas. The Forty-Third International Conference on Machine Learning, 2026.

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|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20<br>[ICML'26]                                                          | CoRe: Collaborative Reasoning via Cross Teaching<br>Kshitij Mishra, Mirat Aubakirov, Martin Takáč, <b>Nils Lukas</b> , Salem Lahlou. The Forty-Third International Conference on Machine Learning, 2026.                                                                                                                                               |
| 19<br>[ACL'26]<br>Main Conference                                        | Forest Before Trees: Latent Superposition for Efficient Visual Reasoning<br>Yubo Wang, Juntian Zhang, Yichen Wu, Yankai Lin, <b>Nils Lukas</b> , Yuhan Liu. The 64th Annual Meeting of the Association for Computational Linguistics, 2026.                                                                                                            |
| 18<br>[ACL'26]<br>Findings                                               | Detecting AI-Generated Video: A Vision-Language Dual-View Survey<br>Dylan Xinming Hou, Juntian Zhang, Xu Gu, Yichen Wu, <b>Nils Lukas</b> , Gus Xia, Xiuying Chen, Yuhan Liu. Findings of the Association for Computational Linguistics, 2026.                                                                                                         |
| 17<br>[ICLR'26]                                                          | DP-Fusion: Token-Level Differentially Private Inference for Large Language Models<br>Rushil Thareja, Preslav Nakov, Praneeth Vepakomma, <b>Nils Lukas</b> . The Fourteenth International Conference on Learning Representations, 2026.                                                                                                                 |
| 16<br>[EACL'26]<br>Findings                                              | SD-E <sup>2</sup> : Semantic Exploration for Reasoning Under Token Budgets<br>Kshitij Mishra, <b>Nils Lukas</b> , Salem Lahlou. Findings of the European Chapter of the Association for Computational Linguistics, 2026.                                                                                                                               |
| 15<br>[NeurIPS'25]<br>AR: 24.5%<br>(5,290/21,575)                        | Mask Image Watermarking<br>Runyi Hu, Jie Zhang, Shiqian Zhao, <b>Nils Lukas</b> , Jiwei Li, Qing Guo, Han Qiu, Tianwei Zhang. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025.                                                                                                                                       |
| 14<br>[EMNLP'25]<br>AR: 22.2%<br>(1,811/8,172)                           | SPIRIT: Patching Speech Language Models against Jailbreak Attacks<br>Amir Djanibekov, Nurdaulet Mukhituly, Kentaro Inui, Hanan Aldarmaki, <b>Nils Lukas</b> . Empirical Methods in Natural Language Processing (Main Conference), 2025.                                                                                                                |
| 13<br>[ICML'25]<br>AR: 26.9%<br>(3,260/12,107)<br>★ Spotlight (Top 2.6%) | Optimizing Adaptive Attacks against Content Watermarks for Language Models<br>Abdulrahman Diaa, Toluwani Aremu, <b>Nils Lukas</b> . The Forty-Second International Conference on Machine Learning, 2025.                                                                                                                                               |
| 12<br>[ICML'25]<br>AR: 26.9%<br>(3,260/12,107)                           | Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems<br>Yutong Wu, Jie Zhang, Yiming Li, Chao Zhang, Qing Guo, Han Qiu, <b>Nils Lukas</b> , Tianwei Zhang. The Forty-Second International Conference on Machine Learning, 2025.                                                                                                               |
| 11<br>[USENIX'24]<br>AR: 19.1%<br>(417/2,176)                            | PEPSI: Practically Efficient Private Set Intersection in the Unbalanced Setting<br>Rasoul Mahdavi, <b>Nils Lukas</b> , Faezeh Ebrahimianghazani, Thomas Humphries, Bailey Kacsmar, John Premkumar, Xinda Li, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.                                               |
| 10<br>[USENIX'24]<br>AR: 19.1%<br>(417/2,176)                            | Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions<br>Abdulrahman Diaa, Lucas Fenaux, Thomas Humphries, Marian Dietz, Faezeh Ebrahimianghazani, Bailey Kacsmar, Xinda Li, <b>Nils Lukas</b> , Rasoul Akhavan Mahdavi, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024. |

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|----------------------|---|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Journal Publications | 9 | <b>[ICLR'24]</b><br>AR: 30.8%<br>(2,250/7,262)                                                                  | <a href="#">Leveraging Optimization for Adaptive Attacks on Image Watermarks</a><br><b>Nils Lukas</b> , Abdulrahman Diaa, Lucas Fenaux, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.                                                                                             |
|                      | 8 | <b>[ICLR'24]</b><br>AR: 30.8%<br>(2,250/7,262)<br>● <b>Media Coverage</b>                                       | <a href="#">Universal Backdoor Attacks</a><br>Benjamin Schneider, <b>Nils Lukas</b> , Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.                                                                                                                                               |
|                      | 7 | <b>[USENIX'23]</b><br>AR: 29.2%<br>(422/1,444)                                                                  | <a href="#">PTW: Pivotal Tuning Watermarking for Pre-Trained Image Generators</a><br><b>Nils Lukas</b> and Florian Kerschbaum. In the 32nd USENIX Security Symposium, 2023.                                                                                                                                                        |
|                      | 6 | <b>[S&amp;P'23]</b><br>AR: 17.0%<br>(195/1,147)<br>★ <b>Distinguished Contribution Award at Microsoft MLADS</b> | <a href="#">Analyzing Leakage of Personally Identifiable Information in Language Models</a><br><b>Nils Lukas</b> , Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, Santiago Zanella-Béguelin. In the 44th IEEE Symposium on Security and Privacy, 2023.                                                                    |
|                      | 5 | <b>[S&amp;P'22]</b><br>AR: 14.5%<br>(147/1,012)                                                                 | <a href="#">SoK: How Robust is Image Classification Deep Neural Network Watermarking?</a><br><b>Nils Lukas</b> , Edward Jiang, Xinda Li, Florian Kerschbaum. In the 43rd IEEE Symposium on Security and Privacy, 2022.                                                                                                             |
|                      | 4 | <b>[ICLR'21]</b><br>AR: 28.7%<br>(860/2,997)<br>★ <b>Spotlight (Top 5%)</b>                                     | <a href="#">Deep Neural Network Fingerprinting by Conferrable Adversarial Examples</a><br><b>Nils Lukas</b> , Yuxuan Zhang, Florian Kerschbaum. The Ninth International Conference on Learning Representations, 2021.                                                                                                              |
|                      | 3 | <b>[IH&amp;MMSEC'21]</b><br>AR: 40.3% (128/318)                                                                 | <a href="#">On the Robustness of Backdoor-based Watermarking in Deep Neural Networks</a><br>Masoumeh Shafieinejad, <b>Nils Lukas</b> , Jiaqi Wang, Xinda Li, Florian Kerschbaum. Proceedings of the 2021 ACM Workshop on Information Hiding and Multimedia Security, 2021.                                                         |
|                      | 2 | <b>[ACSAC'20]</b><br>AR: 20.9% (104/497)                                                                        | <a href="#">Practical Over-Threshold Multi-Party Private Set Intersection</a><br>Rasoul Mahdavi, Thomas Humphries, Bailey Kacsmar, Simeon Krastnikov, <b>Nils Lukas</b> , John Premkumar, Masoumeh Shafieinejad, Simon Oya, Florian Kerschbaum, Erik-Oliver Blass. Annual Computer Security Applications Conference (ACSAC), 2020. |
|                      | 1 | <b>[EuroS&amp;P'20]</b><br>AR: 20.9% (39/187)                                                                   | <a href="#">Differentially Private Two-Party Set Operations</a><br>Bailey Kacsmar, Basit Khurram, <b>Nils Lukas</b> , Alexander Norton, Masoumeh Shafieinejad, Zhiwei Shang, Yaser Baseri, Maryam Sepehri, Simon Oya, Florian Kerschbaum. IEEE European Symposium on Security and Privacy (EuroS&P), 2020.                         |
|                      | 2 | <b>[IEEE S&amp;P'23]</b>                                                                                        | <a href="#">Privacy-Preserving Machine Learning [Cryptography]</a><br>Florian Kerschbaum, <b>Nils Lukas</b> . IEEE Security & Privacy, Volume 21, Issue 6, 2023.                                                                                                                                                                   |
|                      | 1 | <b>[AIP'18]</b>                                                                                                 | <a href="#">SunFlower: A new Solar Tower Simulation Method for use in Field Layout Optimization</a><br>Pascal Richter, Gregor Heimig, <b>Nils Lukas</b> , Martin Frank. AIP Conference Proceedings, Volume 2033, Issue 1, 2018.                                                                                                    |

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|-----------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Workshop Papers | 3<br>[GenAI4Health]                                                       | Sanitizing Medical Documents with Differential Privacy using Large Language Models<br>Rushil Thareja, Gautam Gupta, Preslav Nakov, Praneeth Vepakomma, <b>Nils Lukas</b> . 2025.                                                                    |
|                 | 2<br>[WMARK'25]                                                           | First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge<br>Fahad Shamshad, Tameem Bakr, Yahia Salaheldin Shaaban, Noor Hazim Hussein, Karthik Nandakumar and <b>Nils Lukas</b> . The 1st Workshop on GenAI Watermarking, 2025.   |
|                 | 1<br>[WMARK'25]<br>★ Oral Presentation                                    | Optimizing Adaptive Attacks against Content Watermarks for Language Models<br>Abdulrahman Diaa, Toluwani Aremu, <b>Nils Lukas</b> . The 1st Workshop on GenAI Watermarking, 2025.                                                                   |
| Working Papers  |                                                                           | Mitigating Watermark Forgery in Generative Models via Randomized Key Selection, Submitted.<br>Toluwani Aremu, Noor Hazim Hussein, Munachiso S Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Zhenqiang Gong, <b>Nils Lukas</b> . 2025. |
|                 |                                                                           | Robust and Calibrated Detection of Authentic Multimedia Content, Submitted.<br>Sarim Hashmi, Abdelrahman Elsayed, Mohammed Talha Alam, Samuele Poppi, <b>Nils Lukas</b> . 2025.                                                                     |
| Research Talks  | <b>Adaptively Robust and Forgery-Resistant Watermarking</b>               |                                                                                                                                                                                                                                                     |
|                 | • Meta (FAIR), hosted by <a href="#">Hady Elsahar</a>                     | 2025                                                                                                                                                                                                                                                |
|                 | <b>Optimizing Adaptive Attacks against Content Watermarks</b>             |                                                                                                                                                                                                                                                     |
|                 | • DeepMind, hosted by David Stutz                                         | 2024                                                                                                                                                                                                                                                |
|                 | • University of California, Berkeley, hosted by <a href="#">Dawn Song</a> | 2024                                                                                                                                                                                                                                                |
|                 | <b>Analyzing Leakage of Personal Information in Language Models</b>       |                                                                                                                                                                                                                                                     |
| Keynotes        | • Microsoft M365, hosted by Robert Sim                                    | 2024                                                                                                                                                                                                                                                |
|                 | • Meta, hosted by Will Bullock                                            | 2023                                                                                                                                                                                                                                                |
|                 | • MongoDB, hosted by Marilyn George and Archita Agarwal                   | 2023                                                                                                                                                                                                                                                |
|                 | <b>How Reliable is Watermarking for Image Generators?</b>                 |                                                                                                                                                                                                                                                     |
|                 | • Google, hosted by Somesh Jha                                            | 2023                                                                                                                                                                                                                                                |
|                 | • University of California, Berkeley, hosted by <a href="#">Dawn Song</a> | 2023                                                                                                                                                                                                                                                |
| Work Experience | <b>Aviation Future Week</b> , hosted by Emirates, Dubai                   |                                                                                                                                                                                                                                                     |
|                 | <b>Cyber Energy Leadership Forum</b> , Abu Dhabi                          |                                                                                                                                                                                                                                                     |
| Teaching        | <b>Assistant Professor</b> , MBZUAI, Abu Dhabi, UAE                       |                                                                                                                                                                                                                                                     |
|                 | <b>Research Intern</b> , Royal Bank of Canada, Borealis AI, Toronto       |                                                                                                                                                                                                                                                     |
|                 | • Vertical Federated Learning, hosted by Kevin Wilson                     | since 08/2024                                                                                                                                                                                                                                       |
|                 | <b>Research Intern</b> , Microsoft Research, Cambridge, UK                |                                                                                                                                                                                                                                                     |
|                 | • Privacy for Language Models, hosted by Shruti Tople & Lukas Wutschitz   | 2024                                                                                                                                                                                                                                                |
|                 | <b>Research Assistant</b> , RWTH-Aachen, Aachen                           |                                                                                                                                                                                                                                                     |
| Teaching        | <b>Student Researcher</b> , DSA Daten- und Systemtechnik GmbH, Aachen     |                                                                                                                                                                                                                                                     |
|                 | <b>Software Engineer Intern</b> , A.R. Bayer DSP Systeme GmbH, Düsseldorf |                                                                                                                                                                                                                                                     |
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| Teaching        | <b>Instructor, MBZUAI, UAE</b>                                            |                                                                                                                                                                                                                                                     |
|                 | • ML8509: Collaborative Machine Learning (w/ S. Horvath)                  | 2026                                                                                                                                                                                                                                                |
|                 | • ML8502: Machine Learning Security (14 weeks)                            | 2025                                                                                                                                                                                                                                                |
|                 | • ML807: Federated Learning (7 weeks)                                     | 2025                                                                                                                                                                                                                                                |
|                 | • ML818: Emerging Topics in Trustworthy Machine Learning (4 weeks)        | 2024                                                                                                                                                                                                                                                |

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|-----------------------|------------------------------------------------------------------------------------------------------------------|------------|
| Service               | <b>Teaching Assistant, University of Waterloo, Canada</b>                                                        |            |
|                       | • CS458/658: Computer Security and Privacy                                                                       | 2020, 2021 |
|                       | • CS246 – Object Oriented Programming                                                                            | 2021       |
|                       | <b>Co-Instructor, RWTH-Aachen, Germany</b>                                                                       |            |
|                       | • Course: Data-driven Medicine                                                                                   | 2018       |
|                       | <b>Area Chair</b>                                                                                                |            |
|                       | • International Conference on Learning Representations (ICLR)                                                    | 2026       |
|                       | • International Conference on Machine Learning (ICML)                                                            | 2026       |
|                       | <b>Program Committee</b>                                                                                         |            |
|                       | • ACM Conference on Computer and Communications Security (CCS)                                                   | 2025       |
|                       | • IEEE Symposium on Security and Privacy (IEEE S&P)                                                              | 2025, 2026 |
|                       | • Recent Advances in Intrusion Detection (RAID)                                                                  | 2024       |
|                       | <b>Artifact Evaluation Committee</b>                                                                             |            |
|                       | • The ACM Conference on Computer and Communications Security (CCS)                                               | 2023, 2024 |
|                       | <b>Reviewer</b>                                                                                                  |            |
| PostDocs,<br>Students | • NETYS                                                                                                          | 2025       |
|                       | • ACM TheWebConf (WWW)                                                                                           | 2025       |
|                       | • International Conference on Learning Representations (ICLR)                                                    | 2024, 2025 |
|                       | • International World Wide Web Conference (TheWebConf)                                                           | 2024       |
|                       | • Recent Advances in Intrusion Detection (RAID)                                                                  | 2023       |
|                       | • Neural Information Processing Systems (NeurIPS)                                                                | 2022, 2023 |
|                       | • International Conference on Machine Learning (ICML)                                                            | 2022, 2025 |
|                       | • The Conference on Information and Knowledge Management (CIKM)                                                  | 2020       |
|                       | <b>Other</b>                                                                                                     |            |
|                       | • Sub-Reviewer, Proceedings on Privacy Enhancing Technologies (PETS)                                             | 2021–2023  |
|                       | • Session Chair, IEEE Symposium on Security and Privacy (S&P)                                                    | 2023       |
|                       | • Organization, Workshop on Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA)         | 2018       |
|                       | • Chair for the invited faculty talk program, International Symposium on Trustworthy Foundation Models at MBZUAI | 2025       |
|                       | • Faculty Search Committee, Machine Learning Department at MBZUAI                                                | 2025       |
|                       | • Admission's Committee, MBZUAI Machine Learning Department                                                      | 2025       |
|                       | <b>Student Board Member, Cybersecurity and Privacy Institute</b>                                                 | 2022–2024  |
|                       | <b>School Advisory Committee on Appointments Liaison, CrySP Lab</b>                                              | 2022       |
|                       | <b>Postdoctoral Researchers</b>                                                                                  |            |
|                       | • Samuele Poppi <i>[primary]</i>                                                                                 | since 2025 |
|                       | • Kshitij Mishra <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)                                             | since 2025 |
|                       | • Dushyant Chauhan <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)                                           | since 2025 |
|                       | • Yuhan Liu <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)                                                  | since 2025 |
|                       | <b>PhD Students</b>                                                                                              |            |
|                       | • Fabian Zhang (ML)                                                                                              | since 2025 |
|                       | • Tair Djanibekov (NLP)                                                                                          | since 2025 |
|                       | • Rushil Thareja (NLP) <i>[primary]</i> (w/ Praneeth Vepakomma)                                                  | since 2024 |
|                       | • Toluwani Aremu (ML)                                                                                            | since 2023 |
|                       | <b>MSc Students</b>                                                                                              |            |
|                       | • Juan Nicolas Sepulveda Arias (CV)                                                                              | since 2025 |
|                       | • Mohamed Mahmoud Mohamed Hendy (ML)                                                                             | since 2025 |
|                       | • Majid Mohamed Mohamed Sulaiman Ibrahim (ML)                                                                    | since 2025 |
|                       | • Prince Jha (ML) <i>[secondary]</i> (w/ Salem Lahlou)                                                           | since 2025 |

- Muhammad Mohideen (CV) (*w/ Samuel Horvath*) since 2025

**Alumni**

- Tameem Bakr (ML) MSc · Mar 2026
- Shaikha Jasem Mohamed Ismail Alhosani (ML) MSc · Mar 2026
- Maryam Abdulla Rashed Abdulla Alshamsi (ML) MSc · Mar 2026
- Maryam Alshehyari (ML) MSc · Mar 2026
- Abdalla Khalid Mohamed Abdalla Alzaabi (ML) MSc · Mar 2026
- Ali Al-Ali (ML) MSc · Mar 2026